

Thalassemia Research: Scoring Schemas Documentation

This document outlines the exact scoring logic and schemas used to evaluate participant knowledge and attitudes across all analysis phases.

1. Knowledge Scoring Schemas

Nine single-choice questions and three multiple-choice questions (Q16, Q27, Q28) were evaluated.

A. Phase 1: Raw Knowledge Scoring

This is the simplest metric, treating all facts equally without penalty. *

Single-Choice Correct Answer: +1 point * **Multiple-Choice Correct**

Option Selected: +1 point * **Incorrect, “I don’t know”, or Blank:** 0 points

B. Phase 2: Weighted Knowledge Scoring ($1 - p$ Inverse Difficulty)

This method rewards participants for knowing statistically rare facts. p represents the proportion of the entire cohort that answered the question correctly. * **Single-Choice Correct Answer:** $+ (1 - p)$ points * **Multiple-Choice Correct Option Selected:** $+ (1 - p)$ points * **Incorrect, “I don’t know”, or Blank:** 0 points

C. Phase 3: Positive/Negative Knowledge Scoring

This method actively penalizes dangerous medical misconceptions to account for guessing. * **Single-Choice Correct Answer:** +1 point *

Multiple-Choice Correct Option Selected: +1 point * **Single-Choice**

Incorrect Answer: -1 point (*Penalty*) * **Multiple-Choice Incorrect Option**

Selected: -1 point (*Penalty*) * **“I don’t know”, “Not sure”, or Blank:** 0 points (*No penalty for acknowledging lack of knowledge*)

Correct Knowledge Key Used: * Q15: Yes * Q17: Thalassemia major (severe form) * Q19: Yes * Q20: Very difficult (e.g., bone marrow transplant) * Q21: Can be prevented * Q22: From generation to generation (hereditary) * Q23: Healthy * Q24: Has a chance to be affected (e.g., 25%) * Q26: 40–100

(For multiple-choice questions 16, 27, and 28, all clinically accurate symptoms/actions were scored positively. Options like “Frequent nosebleeds”, “Ignore it”, or “I don’t know” were treated as Incorrect or Neutral respectively).

2. Attitude Scoring Schemas

Attitudes were divided into two categories and scored using both Raw (+1) and Weighted ($1 - p$) methods.

A. Partner Selection Attitudes (Max Raw Score = 3)

- **Q30. Are you willing to have a consanguineous marriage?**
 - *Positive Answer:* Definitely not
- **Q31. Do you accept marriage between two thalassemia carriers?**
 - *Positive Answer:* No
- **Q32. How important is thalassemia screening before marriage?**
 - *Positive Answer:* Very important or Important

B. Cascade Screening Attitudes (Max Raw Score = 3)

- **Q35. Do you think it is important for your family members to undergo screening?**
 - *Positive Answer:* Agree
- **Q39. How easy is it to convince relatives to undergo screening?**
 - *Positive Answer:* Very easy or Somewhat easy
- **Q40. How important is cascade screening in thalassemia prevention?**
 - *Positive Answer:* Very important or Important

Attitude Scoring Logic: * **Raw Method:** +1 point for each positive answer given above. * **Weighted Method:** + $(1 - p)$ points for each positive answer given above (rewarding participants who held positive attitudes that were less common in the cohort).